

"Detailed Functional Requirements"

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1. Introduction & Scope

This document defines the functional requirements for IdeaDNA — an AI-powered Idea Evolution Platform. It specifies what the system must do, how users interact with it, and what outputs are expected. Each requirement includes: ID, Description, Priority, Acceptance Criteria, and Dependencies.

Scope: All 8 core features, authentication, user management, project persistence, and export functionality.
Out of scope: Team matching, investor modules, marketplace (Phase 2).

2. Functional Requirements by Module

Module 1: Authentication & User Management

ID	Requirement	Priority	Acceptance Criteria
FR-AUTH-001	User can register via email/password	HIGH	User receives verification email, account created in DB
FR-AUTH-002	User can register via Google OAuth	HIGH	One-click signup, profile auto-populated
FR-AUTH-003	User can register via GitHub OAuth	HIGH	One-click signup, GitHub profile linked
FR-AUTH-004	User can login with email/password	HIGH	JWT token issued, session established
FR-AUTH-005	User can login with OAuth providers	HIGH	Redirect to provider, callback handling
FR-AUTH-006	User can reset password	HIGH	Email with reset link, token expiry in 1 hour
FR-AUTH-007	User can logout	HIGH	Session invalidated, cookies cleared
FR-AUTH-008	User can view and edit profile	MEDIUM	All fields editable, changes saved immediately
FR-AUTH-009	User can delete account	MEDIUM	Confirmation required, all data deleted within 30 days

Module 2: Idea Generator

ID	Requirement	Priority	Acceptance Criteria
FR-GEN-001	User can select platform category	HIGH	8 platforms available: Web, Mobile, AI/ML, Blockchain, IoT
FR-GEN-002	User can input problem description (10-2000 characters)	HIGH	Text area with character counter, validation
FR-GEN-003	System retrieves matching ideas from curated DB	HIGH	Semantic search returns top 10 matches
FR-GEN-004	AI upgrades retrieved ideas based on user profile	HIGH	Tech stack, complexity, features adjusted
FR-GEN-005	System generates 5-8 idea cards	HIGH	Each card: title, description, scores, tags, difficulty
FR-GEN-006	Cards display innovation score (0-100)	HIGH	Color-coded: green 80+, yellow 60-79, red <60
FR-GEN-007	Cards display feasibility score (0-100)	HIGH	Same color coding as innovation score
FR-GEN-008	User can select ONE idea card	HIGH	Selection triggers lazy-loaded deep-dive
FR-GEN-009	Unselected cards remain accessible	MEDIUM	No deep-dive generated, text-only view
FR-GEN-010	User can regenerate ideas with same input	MEDIUM	New set of ideas, cached results cleared

Module 3: Evolution Engine

ID	Requirement	Priority	Acceptance Criteria
FR-EVO-001	User can input existing idea (title + description)	HIGH	Minimum 20 chars, validation enforced
FR-EVO-002	System runs CriticAgent analysis	HIGH	Weaknesses, gaps, risks, opportunities identified
FR-EVO-003	Critic report visible to user	HIGH	Structured list with severity ratings
FR-EVO-004	System generates 4-6 evolved versions	HIGH	Each with mutation tag, enhancements, fixes
FR-EVO-005	Versions displayed as interactive tree	HIGH	React Flow: root = original, branches = evolved
FR-EVO-006	Tree nodes clickable for details	HIGH	Modal/sidebar with full version info
FR-EVO-007	Node colors based on overall score	HIGH	Green 80+, Yellow 60-79, Red <60
FR-EVO-008	User can select ONE evolved version	HIGH	Triggers same deep-dive as Generator
FR-EVO-009	User can evolve selected version further	MEDIUM	Recursive evolution, new tree generated
FR-EVO-010	Evolution lineage tracked in DB	MEDIUM	Parent-child relations, full history

Module 4: Code Archaeology

ID	Requirement	Priority	Acceptance Criteria
FR-ARC-001	User can paste raw code snippet	HIGH	Text area, max 10000 chars, syntax detection
FR-ARC-002	User can input GitHub repo URL	HIGH	URL validation, public repo access
FR-ARC-003	User can input live website URL	HIGH	URL validation, scraping permission check
FR-ARC-004	System parses code structure	HIGH	Languages, frameworks, components detected
FR-ARC-005	System identifies design patterns	HIGH	Pattern list with confidence scores
FR-ARC-006	System reconstructs evolution timeline	HIGH	v1, v2, v3 with dates and reasons
FR-ARC-007	System infers developer thinking	HIGH	Trade-offs, priorities, constraints, regrets
FR-ARC-008	System generates innovation score	HIGH	0-100 with justification
FR-ARC-009	System suggests future versions	HIGH	3-4 evolved versions based on analysis
FR-ARC-010	Full report exportable as PDF	MEDIUM	Styled PDF with all sections

Module 5: Blueprint Generator (Lazy-Loaded)

ID	Requirement	Priority	Acceptance Criteria
FR-BLU-001	Triggered ONLY on idea selection	HIGH	No pre-generation, on-demand only
FR-BLU-002	Generates Project Overview section	HIGH	2-3 sentence elevator pitch
FR-BLU-003	Generates PRD section	HIGH	Objectives, features, user stories, acceptance criteria
FR-BLU-004	Generates Tech Stack section	HIGH	Frontend, backend, database, AI, devops with reasons
FR-BLU-005	Generates Database Schema section	HIGH	Tables, fields, types, relations, indexes
FR-BLU-006	Generates API Design section	HIGH	Endpoints, methods, payloads, auth, errors
FR-BLU-007	Generates User Flow section	HIGH	Step-by-step with screens and actions
FR-BLU-008	Generates Implementation Plan section	HIGH	Phases, milestones, risks, timeline
FR-BLU-009	All sections personalized to user profile	HIGH	Same idea = different blueprint per user
FR-BLU-010	Exportable as Markdown	HIGH	GitHub README ready format
FR-BLU-011	Exportable as PDF	MEDIUM	Styled PDF with cover page and TOC
FR-BLU-012	Exportable as OpenAPI spec	MEDIUM	YAML format for Swagger UI

Module 6: UI Mockup Generator (Lazy-Loaded)

ID	Requirement	Priority	Acceptance Criteria
FR-MOCK-001	Triggered ONLY on idea selection	HIGH	No pre-generation, on-demand only
FR-MOCK-002	Generates complete HTML file	HIGH	Tailwind CSS via CDN, copy-paste ready
FR-MOCK-003	Includes all required sections	HIGH	Header, Hero, Features, How It Works, Testimonials, CTA
FR-MOCK-004	Uses real content (not lorem ipsum)	HIGH	AI-generated relevant text
FR-MOCK-005	Device toggle: Desktop/Tablet/Mobile	HIGH	Iframe width changes: 100%/768px/375px
FR-MOCK-006	Responsive design implemented	HIGH	Mobile-first, breakpoints at 768px and 1024px
FR-MOCK-007	Dark theme default	HIGH	Matches IdeaDNA aesthetic
FR-MOCK-008	Style variants available	MEDIUM	Modern, Minimal, Cyberpunk, Corporate, Playful
FR-MOCK-009	Exportable as HTML file	HIGH	Download .html file
FR-MOCK-010	Exportable as ZIP	MEDIUM	HTML + CSS + assets folder

Module 7: Innovation Scoring Engine

ID	Requirement	Priority	Acceptance Criteria
FR-SCR-001	Evaluates Originality (0-100)	HIGH	Weight: 30%, justification required
FR-SCR-002	Evaluates Technical Feasibility (0-100)	HIGH	Weight: 25%, user-profile aware
FR-SCR-003	Evaluates Market Potential (0-100)	HIGH	Weight: 25%, competition analysis
FR-SCR-004	Evaluates Complexity Match (0-100)	HIGH	Weight: 20%, timeline/team aware
FR-SCR-005	Calculates weighted overall score	HIGH	Formula: (Orig*0.3)+(Fea*0.25)+(Mkt*0.25)+(Cmp*0.2)
FR-SCR-006	Displays radar chart visualization	HIGH	D3.js, 4 axes, animated fill
FR-SCR-007	Scores personalized per user	HIGH	Same idea = different scores
FR-SCR-008	Provides verdict category	HIGH	Exceptional/Strong/Good/Average/Weak

Module 8: Project Persistence

ID	Requirement	Priority	Acceptance Criteria
FR-PER-001	Auto-saves all projects	HIGH	Every action saved to DB
FR-PER-002	User can view all projects in dashboard	HIGH	List view with filters and search
FR-PER-003	User can search projects by name	HIGH	Text search with instant results
FR-PER-004	User can filter projects by mode	HIGH	Generate/Evolve/Analyze tabs
FR-PER-005	User can filter projects by status	MEDIUM	Draft/Selected/Completed
FR-PER-006	System suggests similar past projects	HIGH	Vector search, similarity > 75%
FR-PER-007	User can view evolution lineage	MEDIUM	Tree view of parent-child relations
FR-PER-008	User can export project as ZIP	MEDIUM	All data + assets packaged
FR-PER-009	User can share project via link	MEDIUM	Public/private toggle
FR-PER-010	User can delete project	HIGH	Confirmation modal, soft delete

3. User Interface Specifications

3.1 Design System

- Color Palette: Primary #1a1a2e, Secondary #16213e, Accent #e94560, Success #00d9ff, Background #0f0f23
- Typography: Inter (headings + body), JetBrains Mono (code). Scale: 28px Hero, 18px H1, 14px H2, 12px Body, 10px Caption
- Spacing: 4px base unit. Margins: 16px small, 24px medium, 32px large, 48px xlarge
- Border Radius: 8px cards, 12px buttons, 16px modals, 9999px pills
- Shadows: sm (0 1px 2px), md (0 4px 6px), lg (0 10px 15px), glow (0 0 20px accent)

3.2 Responsive Breakpoints

- Mobile: 320px - 767px. Single column, stacked layout, hamburger menu, touch targets min 44px
- Tablet: 768px - 1023px. 2-column grid, side navigation, medium images
- Desktop: 1024px+. Full layout, 3-4 column grid, horizontal nav, large images, side panels

4. Data Flow Specifications

All data flows are unidirectional: User Action → API Request → Business Logic → Database → Response → UI Update. React Query handles server state with automatic caching and refetching. Zustand handles client state.

5. Error Handling & Validation

Validation Errors (400)

Zod schema violations. UI shows field-level errors with red borders and messages.

Auth Errors (401/403)

Invalid/missing token or insufficient permissions. Redirect to login or show unauthorized page.

Rate Limit (429)

Too many requests. Show countdown timer, suggest upgrade.

Server Errors (500)

Internal server error. Show generic message, log details, offer retry.

AI API Errors (502/503)

OpenAI service unavailable. Show cached fallback or retry button.

Network Errors

Connection lost. Show offline indicator, queue actions for sync.

6. Integration Requirements

OpenAI API

GPT-4 for generation, text-embedding-3-small for embeddings. Rate limits: Tier-based. Fallback: Cached responses.

GitHub API

REST API v3 for repo analysis. OAuth for private repos. Rate limit: 5000 requests/hour.

Clerk Auth

OAuth providers, JWT management, session handling. Webhooks for user events.

Vercel

Hosting, Edge functions, Analytics, Preview deployments.

Railway/Supabase

PostgreSQL hosting, connection pooling, automated backups.

7. Reporting & Analytics

- User Dashboard: Total projects, ideas generated, average score, favorite platform
- Project Stats: Generation time, token usage, cost per project
- System Analytics: MAU, DAU, retention, funnel conversion, API latency
- Admin Dashboard: User growth, revenue, API costs, error rates

8. Global Acceptance Criteria

- All HIGH priority requirements must be implemented and tested
- All API endpoints must return correct status codes and structured responses
- All user flows must complete without errors in Chrome, Firefox, Safari, Edge
- Mobile responsive design must work on iOS Safari and Android Chrome
- AI generation must complete within 15 seconds for deep-dive features
- Lazy loading must reduce API costs by minimum 70%
- Authentication must be secure (JWT, httpOnly cookies, CSRF protection)
- All user data must be isolated — no cross-user data leakage
- Export functionality must produce valid, usable files (Markdown, PDF, HTML)
- Error handling must provide clear user feedback without exposing sensitive info

— *End of FSD* —